

# **ALY 6010 Prob Theory and Intro Stat – Report**

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## **Final Project**

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## Introduction:

In the world of the seafood industry, particularly in the market of abalone, accurate age determination is crucial for effective management and market valuation. Traditionally, abalone age has been determined through labor-intensive methods involving manual counting of growth rings on their shells. However, advancements in data analysis now allow for the prediction of abalone age using readily available physical measurements. This report explores the use of linear regression models to predict abalone age based on attributes such as length, diameter, and rings, with the goal of simplifying the age determination process. The dataset used in this analysis consists of 4176 records, each containing measurements of various physical attributes of abalone, including shell dimensions, body weight, length, and ring count.

## Problem statement:

As age determination through microscopic examination of shell rings is time-consuming and expensive, developing accurate predictive models is crucial. Abalone, a rich and nutritious food resource in many parts of the world, is highly valued economically, with its price positively correlated with age. The objective of this study is to construct linear regression models to predict abalone age based on easily measurable physical characteristics.

## Data Used:

**Reading dataset :** Importing the file on dataset in R and reading dataset

```
abalone <- read.csv("abalone.data")
```

Description: df [6 × 9]

|   | M<br><chr> | X0.455<br><dbl> | X0.365<br><dbl> | X0.095<br><dbl> | X0.514<br><dbl> | X0.2245<br><dbl> | X0.101<br><dbl> | X0.15<br><dbl> | X15<br><int> |
|---|------------|-----------------|-----------------|-----------------|-----------------|------------------|-----------------|----------------|--------------|
| 1 | M          | 0.350           | 0.265           | 0.090           | 0.2255          | 0.0995           | 0.0485          | 0.070          | 7            |
| 2 | F          | 0.530           | 0.420           | 0.135           | 0.6770          | 0.2565           | 0.1415          | 0.210          | 9            |
| 3 | M          | 0.440           | 0.365           | 0.125           | 0.5160          | 0.2155           | 0.1140          | 0.155          | 10           |
| 4 | I          | 0.330           | 0.255           | 0.080           | 0.2050          | 0.0895           | 0.0395          | 0.055          | 7            |
| 5 | I          | 0.425           | 0.300           | 0.095           | 0.3515          | 0.1410           | 0.0775          | 0.120          | 8            |
| 6 | F          | 0.530           | 0.415           | 0.150           | 0.7775          | 0.2370           | 0.1415          | 0.330          | 20           |

6 rows

## Installing required libraries

```
library(readr)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
## filter, lag
```

```
## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union

library(tidyr)
library(ggplot2)
library(caret)

## Loading required package: lattice

library(reshape2)

##
## Attaching package: 'reshape2'

## The following object is masked from 'package:tidyr':
##
## smiths
```

### Explore the structure of the dataset

Structuring the dataset to examine its characteristics, including the variables it contains, their types, the number of observations, and any missing values or outliers.

```
str(abalone)

## 'data.frame':      4176 obs. of  9 variables:
## $ M : chr "M" "F" "M" "I" ...
## $ X0.455 : num 0.35 0.53 0.44 0.33 0.425 0.53 0.545 0.475 0.55 0.525 ...
## $ X0.365 : num 0.265 0.42 0.365 0.255 0.3 0.415 0.425 0.37 0.44 0.38 ...
## $ X0.095 : num 0.09 0.135 0.125 0.08 0.095 0.15 0.125 0.125 0.15 0.14 ..
## $ X0.514 : num 0.226 0.677 0.516 0.205 0.351 ...
## $ X0.2245: num 0.0995 0.2565 0.2155 0.0895 0.141 ...
## $ X0.101 : num 0.0485 0.1415 0.114 0.0395 0.0775 ...
## $ X0.15 : num 0.07 0.21 0.155 0.055 0.12 0.33 0.26 0.165 0.32 0.21 ...
## $ X15 : int 7 9 10 7 8 20 16 9 19 14 ...
```

### Data Types:

M: Character (categorical)

X0.455, X0.365, X0.095, X0.514, X0.2245, X0.101, X0.15: Numeric

X15: Integer

### Observations:

There are 4176 observations in total.

## Data Cleaning and Transformation:

Data cleaning and transformation are crucial steps in the data analysis process, aimed at ensuring that the dataset is accurate, complete, and formatted appropriately for analysis.

### Renaming columns

The dataset's columns have been renamed to enhance clarity and understanding. This involved assigning new names to the various variables in the data. By doing so, it simplifies the process for users to comprehend the meaning of each part of the dataset. These alterations improve clarity and ensure that individuals can interpret and analyze the information without confusion.

```
abalone <- abalone %>%
  rename(Sex = M,
         Length = X0.455,
         Diameter = X0.365,
         Height = X0.095,
         Whole_weight = X0.514,
         Shucked_weight = X0.2245,
         Viscera_weight = X0.101,
         Shell_weight = X0.15,
         Rings = X15)
```

Description: df [6 x 9]

|   | Sex<br><chr> | Length<br><dbl> | Diameter<br><dbl> | Height<br><dbl> | Whole_weight<br><dbl> | Shucked_weight<br><dbl> | Viscera_weight<br><dbl> | Shell_weight<br><dbl> | Rings<br><int> |
|---|--------------|-----------------|-------------------|-----------------|-----------------------|-------------------------|-------------------------|-----------------------|----------------|
| 1 | M            | 0.350           | 0.265             | 0.090           | 0.2255                | 0.0995                  | 0.0485                  | 0.070                 | 7              |
| 2 | F            | 0.530           | 0.420             | 0.135           | 0.6770                | 0.2565                  | 0.1415                  | 0.210                 | 9              |
| 3 | M            | 0.440           | 0.365             | 0.125           | 0.5160                | 0.2155                  | 0.1140                  | 0.155                 | 10             |
| 4 | I            | 0.330           | 0.255             | 0.080           | 0.2050                | 0.0895                  | 0.0395                  | 0.055                 | 7              |
| 5 | I            | 0.425           | 0.300             | 0.095           | 0.3515                | 0.1410                  | 0.0775                  | 0.120                 | 8              |
| 6 | F            | 0.530           | 0.415             | 0.150           | 0.7775                | 0.2370                  | 0.1415                  | 0.330                 | 20             |

6 rows

### Check for missing values

Checking for missing values in a dataset is essential to facilitate data preprocessing and ensure data quality.

```
summary(is.na(abalone))

##      Sex      Length      Diameter      Height
## Mode :logical Mode :logical Mode :logical Mode :logical
## FALSE:4176    FALSE:4176    FALSE:4176    FALSE:4176
## Whole_weight  Shucked_weight Viscera_weight Shell_weight
## Mode :logical Mode :logical Mode :logical Mode :logical
## FALSE:4176    FALSE:4176    FALSE:4176    FALSE:4176
## Rings
```

```
##                                     Mode                                     :logical
## FALSE:4176
```

### identifying class of data set

```
abalone.class <- apply(abalone, 1, FUN = function(x) class(x))
abalone.class
```

| ## | Sex            | Length         | Diameter     | Height    | Whole_weight |
|----|----------------|----------------|--------------|-----------|--------------|
| ## | "character"    | "numeric"      | "numeric"    | "numeric" | "numeric"    |
| ## | Shucked_weight | Viscera_weight | Shell_weight |           | Rings        |
| ## | "numeric"      | "numeric"      | "numeric"    | "integer" |              |

## Explanatory Data Analysis (EDA):

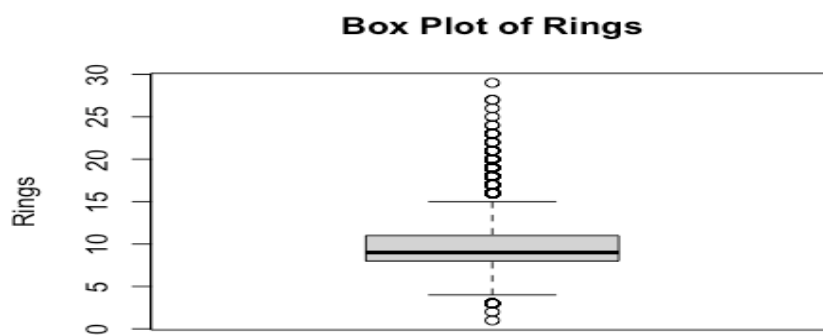
Conducting a thorough EDA to understand the distributions, relationships, and trends within the data. Using visualizations such as scatter plots, histograms, and box plots to examine the relationships between the dependent and independent variables.

### Rings Distribution:

"Rings" variable, it serves as the target variable in the dataset. Each integer value in the "Rings" variable corresponds to the number of rings observed on an abalone's shell. Converting this value to years is achieved by adding 1.5 to the integer value, providing an approximate age in years. Understanding the distribution of the "Rings" variable is crucial for model development and analysis. It helps in identifying the range and spread of ring counts, which are essential factors for predicting abalone age accurately.

### Box Plot of Rings

```
boxplot(abalone$Rings, main = "Box Plot of Rings", ylab = "Rings")
```



**Description:** The box plot provides a visual summary of the central tendency, spread, and potential outliers in the dataset, offering insights into the distribution of abalone rings where the mean ranges between 5-10 rings, with the minimum and maximum number of rings set at 1 and 29, respectively.

## Creating intervals

: Creating intervals for the number of rings in abalone to visualize the distribution of ring counts across the dataset.

```
intervals      <-      c(0,      5,      10,      15,      20,      25,      30)
intervals

## [1]  0  5 10 15 20 25 30

interval_labels <- paste(intervals[-length(intervals)], intervals[-1], sep = "-")
interval_labels

## [1] "0-5"  "5-10" "10-15" "15-20" "20-25" "25-30"
```

## Ring Frequency table:

The output displays a frequency table of the intervals created for the number of rings in abalone. Each interval represents a range of ring counts, and the corresponding frequencies indicate how many abalones fall within each interval.

```
ring_intervals <- cut(abalone$Rings, breaks = intervals, right = FALSE)
frequency_table <-      table(ring_intervals)
print(frequency_table)
```

|    | ring_intervals                               |
|----|----------------------------------------------|
|    | [0,5) [5,10) [10,15) [15,20) [20,25) [25,30) |
| ## | 74 2022 1717 301 57 5                        |

Convert interval\_labels to factor with desired order of levels

```
ring_frequencies <- hist(abalone$Rings, breaks = intervals, plot = FALSE)$counts
```

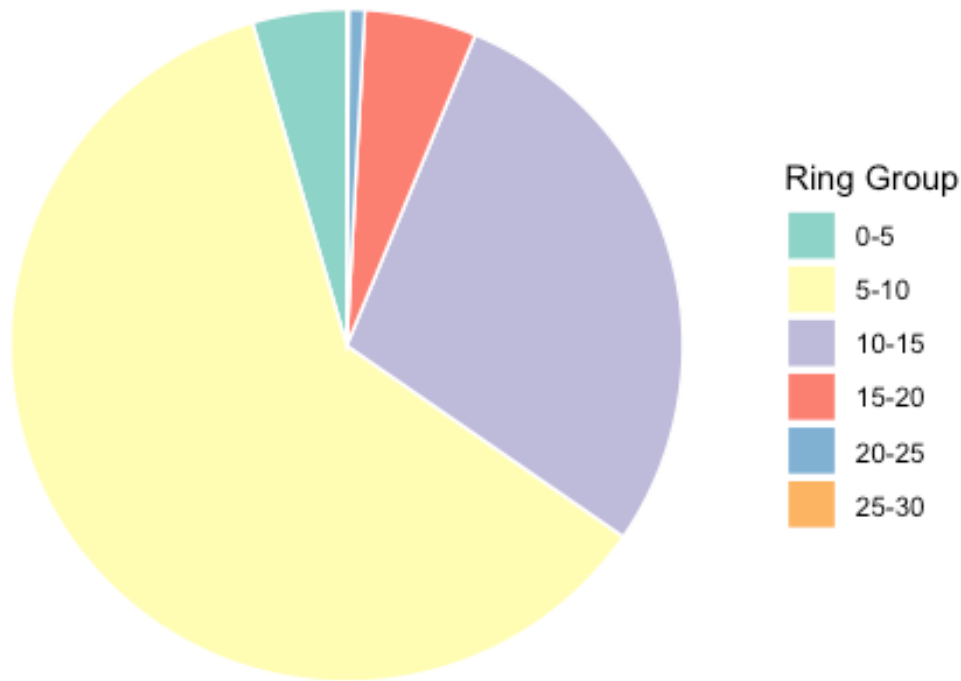
## Pie chart of rings distribution

```
pie_data <- data.frame(interval = interval_labels, frequency = ring_frequencies)
pie_data$interval <- factor(pie_data$interval, levels = interval_labels)
pie_data$interval

## [1] 0-5 5-10 10-15 15-20 20-25 25-30
## Levels: 0-5 5-10 10-15 15-20 20-25 25-30

pie_chart <- ggplot(pie_data, aes(x = "", y = frequency, fill = interval)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar("y", start = 0) +
  scale_fill_brewer(palette = "Set3") +
  theme_void() +
  theme(legend.position = "right") +
  labs(title = "Distribution of Rings in Abalone", fill = "Ring Group")
print(pie_chart)
```

## Distribution of Rings in Abalone



**Description:** The pie chart illustrates the distribution of ring counts among abalones, categorized into intervals. Each interval represents a range of ring counts, and the corresponding frequencies indicate the number of abalones falling within each interval. Through the pie chart, one can visually assess the proportion of abalones falling within each ring count interval, providing a clear depiction of the distribution pattern within the dataset.

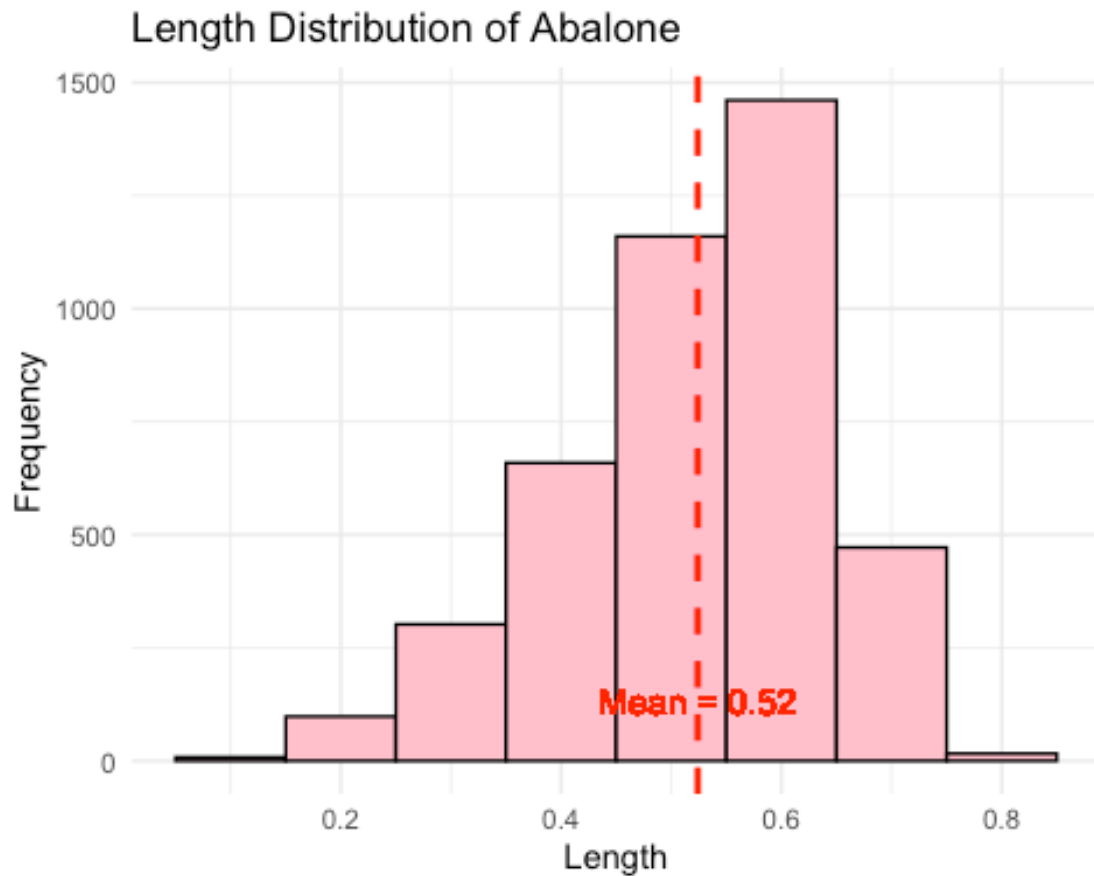
### Mean Distribution length

```
mean_Length <- mean(abalone$Length, na.rm = TRUE)
print(mean_Length)

## [1] 0.5240086

ggplot(abalone, aes(x = Length)) +
  geom_histogram(binwidth = 0.1, fill = "pink", color = "black") +
  geom_vline(xintercept = mean_Length, color = "red", linetype = "dashed", size = 1) +
  geom_text(aes(x = mean_Length, label = paste("Mean =", round(mean_Length, 2))),
  ),
  labs(title = "Length Distribution of Abalone",
  x = "Length",
  y = 50, vjust = -1, color = "red", size = 4) +
  theme_minimal()
```

```
theme_minimal() = "Frequency")
```



**Description:** By visualizing the mean distribution, we can assess the overall variability and dispersion of abalone lengths within each subset, aiding in further analysis and interpretation.

### Diameter Mean

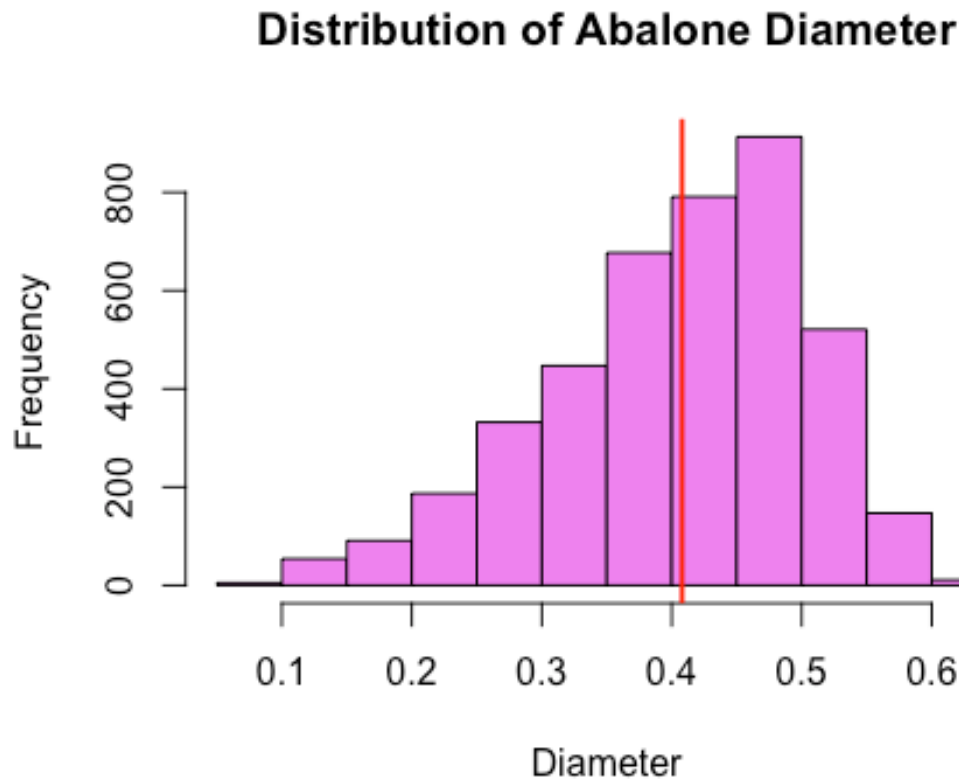
```
mean_Diameter <- mean(abalone$Diameter, na.rm = TRUE)
print(mean_Diameter)
## [1] 0.4078915
```

### Distribution of diameter

```
mean_Diameter <- mean(abalone$Diameter, na.rm = TRUE)
print(mean_Diameter)
## [1] 0.4078915
hist(abalone$Diameter,
     main = "Distribution of Abalone Diameter",
     xlab = "Diameter",
     ylab = "Frequency",
```



```
col = "violet")
abline(v = mean(abalone$Diameter), col = "red", lwd = 2)
```



**Description:** The plot illustrates the distribution of mean(0.4078) diameter of abalones. This distribution showcases the central tendency of diameter measurements across the dataset, providing insights into the typical size of abalones in terms of their diameter.

### Training the dataset

Dataset training allows the model to learn patterns and relationship present in the data.

```
trainingSet = read.csv('abalone.data')
head(trainingSet)
```

| ## |   | M | X0.455 | X0.365 | X0.095 | X0.514 | X0.2245 | X0.101 | X0.15 | X15 |
|----|---|---|--------|--------|--------|--------|---------|--------|-------|-----|
| ## | 1 | M | 0.350  | 0.265  | 0.090  | 0.2255 | 0.0995  | 0.0485 | 0.070 | 7   |
| ## | 2 | F | 0.530  | 0.420  | 0.135  | 0.6770 | 0.2565  | 0.1415 | 0.210 | 9   |
| ## | 3 | M | 0.440  | 0.365  | 0.125  | 0.5160 | 0.2155  | 0.1140 | 0.155 | 10  |
| ## | 4 | I | 0.330  | 0.255  | 0.080  | 0.2050 | 0.0895  | 0.0395 | 0.055 | 7   |
| ## | 5 | I | 0.425  | 0.300  | 0.095  | 0.3515 | 0.1410  | 0.0775 | 0.120 | 8   |
| ## | 6 | F | 0.530  | 0.415  | 0.150  | 0.7775 | 0.2370  | 0.1415 | 0.330 | 20  |

**Correlation coefficient of rings and length of abalone**

```
correlation_coefficient <- cor(abalone$Rings, abalone$Length)
correlation_coefficient

## [1] 0.5571226
```

The **correlation coefficient** between “Rings” and “Length” in the abalone dataset is approximately 0.557. This positive correlation coefficient suggests that there is a moderate positive linear relationship between the number of rings and the length of abalones.

**Correlation coefficient of rings and diameter of abalone**

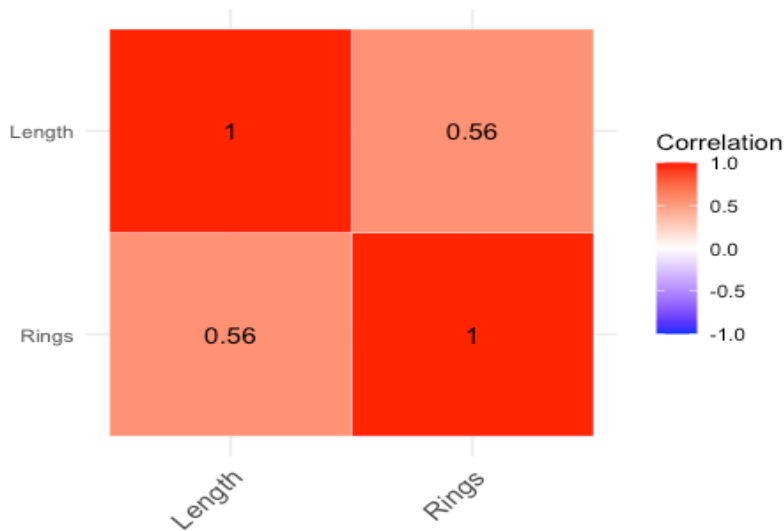
```
correlation_coefficient <- cor(abalone$Rings, abalone$Diameter)
correlation_coefficient

## [1] 0.5750054
```

The **correlation coefficient** between “Rings” and “Diameter” in the abalone dataset is approximately 0.5750. This positive correlation coefficient suggests that there is a moderate positive linear relationship between the number of rings and the diameter of abalones.

**Creating correlation matrix**

```
correlation_matrix <- cor(abalone[c("Rings", "Length")])
correlation_df <- as.data.frame(correlation_matrix)
correlation_df$Var1 <- rownames(correlation_df)
melted_df <- melt(correlation_df, id.vars = "Var1", variable.name = "Var2", value.name = "Correlation")
ggplot(data = melted_df, aes(x = Var1, y = Var2, fill = Correlation)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0,
    limit = c(-1, 1), space = "Lab", name = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, size = 12, hjust = 1)) +
  labs(x = "", y = "") + # Remove axis labels
  geom_text(aes(label = round(Correlation, 2)), color = "black", size = 4) +
  # Add labels
  coord_fixed()
```



**Description:** The correlation matrix between the “Rings” and “Length” of abalone represents the pairwise correlations between these two variables. Each cell in the matrix contains a correlation coefficient that quantifies the strength and direction of the linear relationship between “Rings” and “Length”. The positive (closer to +1), indicates a positive linear relationship, meaning that as one variable (“Length”) increases, the other variable (“Rings”) tends to increase as well.

## Modeling:

Splitting the dataset into training and testing sets to evaluate the performance of the model. Developing a linear regression model using the selected features. Also, considering building a models to compare two sets of features.

```
model <- lm(Rings ~ Length, data = abalone)
model

##
##                                     Call:
##   lm(formula = Rings ~ Length, data = abalone)
##
##                                     Coefficients:
##   (Intercept)                2.097                14.953
summary(model)

##
##                                     Call:
##   lm(formula = Rings ~ Length, data = abalone)
##
##                                     Residuals:
##               Min               1Q               Median               3Q              Max
##   -5.9660    -1.6973    -0.7417     0.8751    16.6789
```

```
##
##                                     Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.0968      0.1855  11.30  <2e-16 ***
## Length          14.9534      0.3450  43.34  <2e-16 ***
##
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.677 on 4174 degrees of freedom
## Multiple R-squared:  0.3104, Adjusted R-squared:  0.3102
## F-statistic: 1879 on 1 and 4174 DF, p-value: < 2.2e-16
```

## MODEL SUMMARY

**Call:** This section shows the call that was used to create the linear regression model. It specifies the formula used ( $\text{Rings} \sim \text{Length}$ ) and the dataset (abalone) that was used for fitting the model.

**Coefficients:** The coefficients section provides the estimated coefficients for the intercept and the predictor variable (Length).

## RESULT OF FITTED MODEL

**Intercept:** The intercept term is estimated to be 2.097. This represents the predicted value of the response variable ("Rings") when the predictor variable ("Length") is zero. However, in practical terms, this interpretation might not make sense depending on the context of the data.

**Length:** The coefficient for the predictor variable "Length" is estimated to be 14.953. This suggests that for each unit increase in the length of the abalone, the predicted number of rings increases by approximately 14.953 units.

**Significance:** Both the intercept and the coefficient for "Length" are highly significant, as indicated by the very small p-values (typically  $< 0.05$ ).

**R-squared:** The R-squared value of 0.3104 indicates that approximately 31.04% of the variance in the number of rings can be explained by the linear relationship with the length of the abalone.

**Adjusted R-squared:** The adjusted R-squared value of 0.3102 adjusts for the number of predictors in the model.

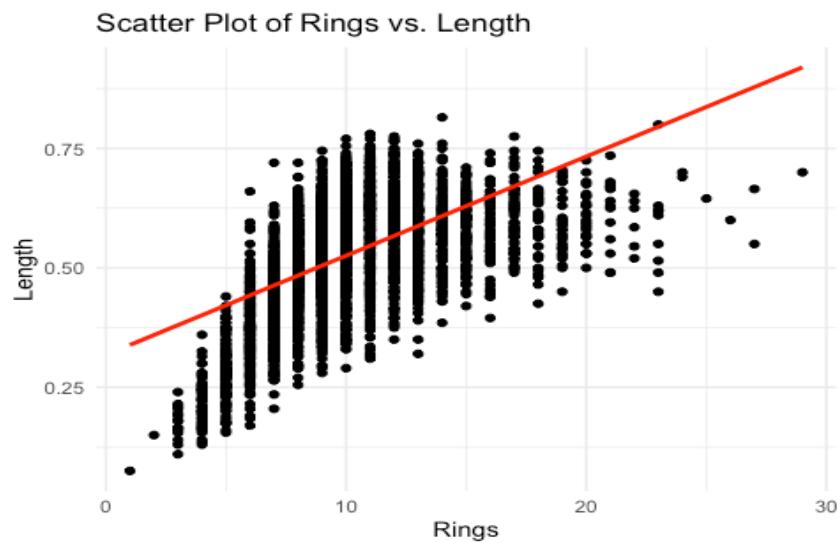
**F-statistic:** The F-statistic of 1879 with an associated very small p-value indicates that the model as a whole is highly significant.

In summary, the linear regression model suggests that the length of the abalone is a significant predictor of the number of rings, and approximately 31% of the variance in the number of rings can be explained by the length of the abalone alone.

### Linear regression plot between rings and length

```
ggplot(abalone, aes(x = Rings, y = Length)) +
  geom_point() + # Add points
  geom_smooth(method = "lm", se = FALSE, color = "red") + # Add linear regression line
  labs(x = "Rings", y = "Length") + # Label axes
  ggtitle("Scatter Plot of Rings vs. Length") + # Add title
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```



**Description:** Scatter plot shows a linear regression line overlaid, visualizing the relationship between the number of rings and the length of abalone specimens. This plot helps in understanding how the length of an abalone relates to the number of rings it possesses, potentially indicating growth patterns or other biological relationships.

**Linear regression model between Diameter and rings of abalone** Linear regression model fitted to predict the diameter of abalones based on the number of rings they have.

```
model <- lm(Diameter ~ Rings, data = abalone)
model

##
##                                     Call:
## lm(formula = Diameter ~ Rings, data = abalone)
##
##                                     Coefficients:
## (Intercept)          0.2321          0.0177
##
summary(model)

##
##                                     Call:
```

```
## lm(formula = Diameter ~ Rings, data = abalone)
##
##                               Residuals:
##      Min       1Q   Median       3Q      Max
## -0.259232 -0.058680  0.005913  0.063616  0.203209
##
##                               Coefficients:
##              (Intercept)  0.2320521  0.0040714  57.00  <2e-16 ***
##              Rings        0.0177035  0.0003899  45.41  <2e-16 ***
##
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08121 on 4174 degrees of freedom
## Multiple R-squared:  0.3306, Adjusted R-squared:  0.3305
##
## F-statistic: 2062 on 1 and 4174 DF, p-value: < 2.2e-16
```

## MODEL SUMMARY

**Call:** This section shows the call that was used to create the linear regression model. It specifies the formula used (Diameter ~ Rings) and the dataset (abalone) that was used for fitting the model.

**Coefficients:** The coefficients section provides the estimated coefficients for the intercept and the predictor variable (Rings).

## RESULT OF FITTED MODEL

The residuals, representing the differences between observed and predicted diameter values, exhibit a spread ranging from -0.259232 to 0.203209 millimeters.

**Intercept (0.2321):** This represents the estimated mean diameter when the number of rings is zero.

**Rings (0.0177):** For each additional ring, the diameter is estimated to increase by approximately 0.0177 millimeters.

**Significance :** Both the intercept and the coefficient for "Rings" are highly significant, as indicated by the very small p-values (typically < 0.05). This suggests that they have a statistically significant impact on the diameter of abalones.

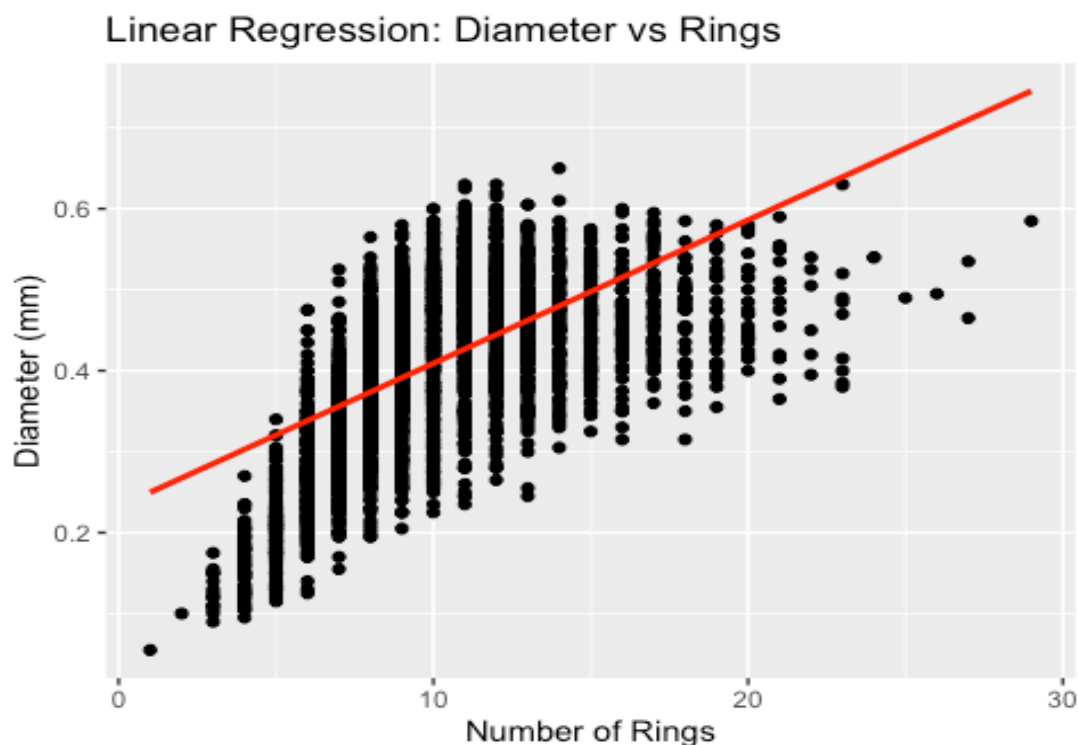
**R-squared Value:** The R-squared value (0.3306) indicates that approximately 33.06% of the variance in the diameter can be explained by the linear relationship with the number of rings.

## F-statistic:

The F-statistic (2062) tests the overall significance of the model. With a very small p-value ( $< 2.2e-16$ ), it indicates that the model as a whole is highly significant, implying that predictor "Rings" has a non-zero effect on the diameter.

## Linear regression plot between rings and diameter

```
ggplot(abalone, aes(x = Rings, y = Diameter)) +
  geom_point() + # Scatter plot of the data points
  geom_smooth(method = "lm", se = FALSE, color = "red") + # Add linear regression line
  labs(title = "Linear Regression: Diameter vs Rings",
        x = "Number of Rings",
        y = "Diameter (mm)")
## `geom_smooth()` using formula = 'y ~ x'
```



**Description:** The scatter plot shows the direction of the relationship between "Rings" and "Diameter." Since the coefficient for "Rings" is positive (0.0177), the regression line has a positive slope, indicating that as the number of rings increases, the diameter tends to increase as well.

## Evaluation and Results:

Evaluating the performance of the model using metrics, such as R-squared, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

Between “RINGS” and “LENGTH”

```

predictions <- predict(model)

# Calculate residuals
residuals <- abalone$Rings - predictions

# Calculate R-squared
rsquared <- summary(model)$r.squared
print(paste("R-squared:", rsquared))

## [1] "R-squared: 0.310385572625062"

# Calculate Mean Absolute Error (MAE)
mae <- mean(abs(residuals))
print(paste("Mean Absolute Error (MAE):", mae))

## [1] "Mean Absolute Error (MAE): 1.95049213480304"

# Calculate Mean Squared Error (MSE)
mse <- mean(residuals^2)
print(paste("Mean Squared Error (MSE):", mse))

## [1] "Mean Squared Error (MSE): 7.16448568172938"

```

### Result driven:

**R-squared ( $R^2$ ):** R-squared measures the proportion of the variance in the dependent variable (Rings) that is predictable from the independent variable (Length) in the model. In this case, R-squared is approximately 0.31, which means that around 31% of the variance in Rings can be explained by the linear relationship with Length.

**Mean Absolute Error (MAE):** MAE measures the average absolute difference between the actual values and the predicted values. Here, MAE is approximately 1.95. It means, on average, the model's predictions are off by around 1.95 units of Rings.

**Mean Squared Error (MSE):** MSE measures the average squared difference between the actual and predicted values. In this case, MSE is approximately 7.16. It's larger than MAE because it penalizes larger errors more heavily due to squaring. Interpretation:

Between “RINGS” and “DIAMETER”

```

# Predictions using the model
predictions <- predict(model, newdata = abalone)

```



```

#                               R-squared
rsquared                        <- summary(model)$r.squared

#                               Mean           Absolute           Error           (MAE)
mae                            <- mean(abs(predictions) - abalone$Diameter))

#                               Mean           Squared           Error           (MSE)
mse                            <- mean((predictions - abalone$Diameter)^2)

#                               Print           the           metrics
print(paste("R-squared:", rsquared))

## [1] "R-squared: 0.330631201896034"

print(paste("Mean Absolute Error (MAE):", mae))

## [1] "Mean Absolute Error (MAE): 0.0671556510764402"

print(paste("Mean Squared Error (MSE):", mse))

## [1] "Mean Squared Error (MSE): 0.0065920179550055"

```

### Result driven:

**R-squared (R<sup>2</sup>):** The R-squared value of approximately 0.331 indicates that around 33.1% of the variance in the diameter can be explained by the linear relationship with the number of rings.

**Mean Absolute Error (MAE):** The Mean Absolute Error (MAE) of approximately 0.067 indicates, on average, the absolute difference between the predicted diameter values and the actual diameter values is 0.067 millimeters. This metric provides a measure of the average magnitude of errors in the predictions.

**Mean Squared Error (MSE):** The Mean Squared Error (MSE) of approximately 0.00659 represents the average of the squared differences between the predicted diameter values and the actual diameter values. It is a measure of the average squared deviation of predictions from the actual values. Since MSE squares the errors, it penalizes larger errors more heavily than smaller errors.

## **Conclusion:**

In conclusion, a linear regression model has been applied to various variables in the "abalone" dataset.

The linear regression model suggests that abalone length is a significant predictor of the number of rings, explaining approximately 31% of the variance in the number of rings. However, it's essential to recognize that this model may have limitations.

While the statistically significant relationship between the number of rings and abalone diameter, as confirmed by the linear regression model, provides valuable insights, it's important to acknowledge the limitations of relying solely on a 33.1% variance explanation in predicting abalone age. A variance explanation of 33.1% indicates that there is still a substantial amount of unexplained variability in the age prediction process.

Given the complexities involved in abalone growth patterns and the multitude of factors that can influence their development, relying solely on variables such as length, diameter or rings may not provide sufficient accuracy in age prediction, especially considering the limitations of having a relatively small dataset. Other factors, such as environmental conditions, food availability, and genetic variability, can also play significant roles in determining abalone age but may not be fully captured by these variables alone.